

professional logistics companies. We are also going to launch an e-commerce platform soon on our website.”

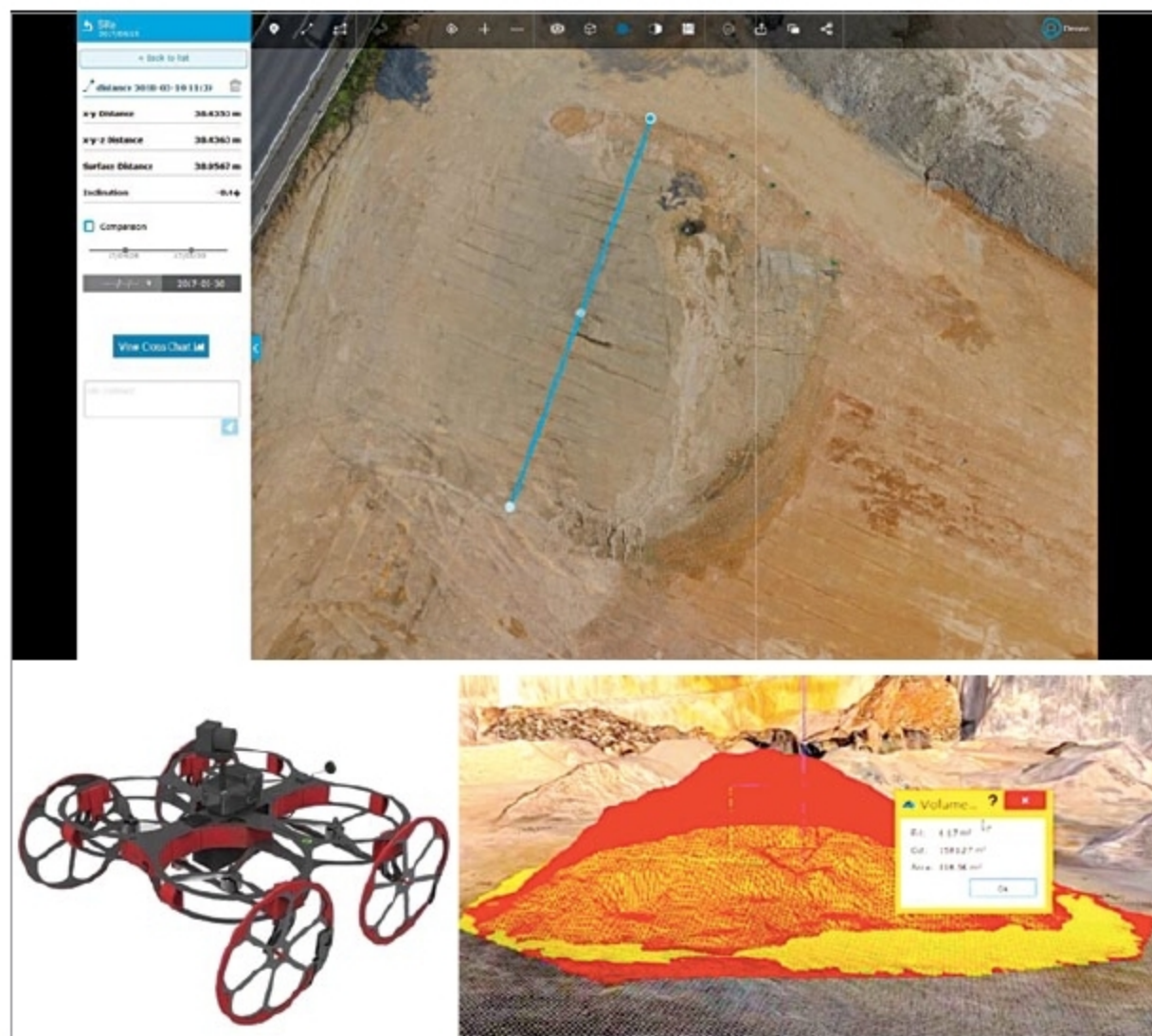
The convenience and eco-friendly features offered give tough competition to petrol- and diesel-run vehicles. These

e-vehicles are being used by customers already. Vadera elaborates, “Etron’s entry-level model is targeted towards individuals about thirteen years of age. Egnite is targeted towards professional bicyclists, Etrance towards teenagers

and women, and Epluto towards a wide-ranging audience.” With two-wheelers already launched, the company plans to partner with others to develop lithium batteries for three- and four-wheelers.

—Ayushee Sharma

2 EXPANSION OF THE UAV ECOSYSTEM IN INDIA



Originally used in military applications, an unmanned aerial vehicle (UAV), or a drone, is today found almost everywhere. According to ‘UAV Payload and Subsystems Market: Global Industry Trends, Share, Size, Growth, Opportunity and Forecast 2019-2024’ report, the market is expected to reach a value of US\$ 11.6 billion by 2024, at a compound annual growth rate (CAGR) of 7.3 per cent during 2019-2024.

Established in 2018 to offer a complete UAV ecosystem, Hyderabad-based Terra Drone India Pvt Ltd provides end-to-end drone solutions based on customer needs. It works mainly on UAV engineering, geographic informa-

tion system (GIS) solutions, videography and software development. It is an associate company of Japan’s Terra Drone Corp. founded in 2016, at a time when the Japanese government was promoting i-Construction initiative under which construction sites would integrate information and communications technology (ICT) to increase productivity.

The startup is completely indigenous, thus boosting the Indian market. Regarding government support, Praatek Srivastava, chief executive officer, Terra Drone India, says, “Terra Drone India has found support from the current government, especially in terms of single-window registration, ease of

doing business and bringing proven foreign technologies to India through initiatives like Startup India and Make in India. Department for Promotion of Industry and Internal Trade (DPIIT) and National Small Industries Corp. (NSIC) have helped startups like us to operate on par with established players in the drone industry. The policy of Telangana state government to support locally-based new-age technology companies such as Terra Drone India has also been very encouraging.”

Hardware products have been designed to cater to the surveying needs of their clients, and include Terra Wing, Terra SLAM and Terra LiDAR (light detection and ranging) system. The range of their operations is 7km. Communication medium for the drones is radio telemetry, which allows the operator to accurately monitor the conditions while flying.

Terra Wing is a dedicated survey-grade, fixed-wing unmanned aerial system (UAS) designed to cover large areas in a short time. It has a 24/42MP camera, 9000mAh battery and a speed range of 10m/s to 25m/s. It finds application in infrastructure inspection, agriculture and so on.

Terra SLAM is a dust- and water-resistant solution built for underground mines. It uses lithium-polymer batteries, and has a flight time of up to twenty minutes.

Terra LiDAR uses pulsed laser light to attain an accuracy of 3cm to 5cm, necessary to provide high-definition 3D models. It finds use in plant height assessment, topographic mapping, digital preservation and the like.

Sharing how the pricing is done, Srivastava says, “Since we are into the services industry, pricing varies from project to project. We also help clients procure drones and related sensors, if